

GRA 2

The exponential function $y = a^x$

Exponential functions are described by a mathematical relationship of the form $y = a^x$. Note that the variable is in the exponent. a is the base and it is raised to the power of the controlling variable, x . We use only positive values for the base.

We are dealing with an exponential function whenever the growth rate of a certain quantity is constant. For instance the growth rate of the human population is approximately 1,53% per year. That means that if the population was p_1 in a particular year it would be $1,0153 p_1$ in the following year. The startling nature of this type of function is that although the growth rate might seem small the quantity itself increases (or decreases) quite fast (exponentially) over time. Examples of such processes are:

- The decay of radioactive material. The amount of radioactive isotopes decreases exponentially with time as they mutate to other elements. In this case the relation is of the form $y = a^{-x}$ to reflect that the quantity is decreasing.
- The growth rate of a colony of bacteria, given enough resources, is directly proportional to the size of the colony at that particular time. That is, the amount by which they change per unit time, or db/dt , becomes more as the amount of bacteria that is present increases, or $db/dt \propto b$. This is evident in the fact that the gradient of the graph of the amount of bacteria present as a function of time increases as the amount of bacteria increases - see **figure 1**. The proportionality constant is the growth rate, r , of the colony (which is constant). We incorporate the growth rate as follows into the exponential function

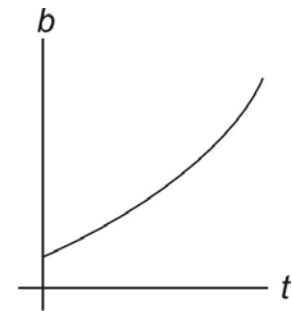


Fig. 1 The amount of bacteria in a colony with time.

$$b = a^{rt}$$

In this way it controls how fast the function is increasing.

- The emptying of water from a tank.
- The reduction in air pressure with height.
- The discharging of a capacitor in a circuit. The rate at which the capacitor discharges decreases as the amount of electric charge on it decreases, similar to the rate at which a colony of bacteria changes.

For naturally occurring processes we often find that the base $e = 2,71828$ should be used. The exponential function then becomes

$$y = y_0 e^{-rx}$$

for decreasing functions and

$$y = y_0 e^{rx}$$

for increasing functions. y_0 is the starting value for the function, that is when the control variable, x , is zero.

Logarithms

If $y = a^x$ then we say that the logarithm of y to the base a is x or $\log_a(y) = x$. The logarithm of a number is therefore the exponent to which the base must be raised in order to obtain that number.

Common logarithms are to the base 10. In such a case the base is not indicated explicitly, for example $\log(100) = 2$. We speak of natural or Napierian logarithms when the base is e . In such a case we write $\ln$ rather than $\log_e$, for example $\ln(2,00) = 0,693$.

Transforming graphs of exponential functions to straight-line graphs.

If one physical quantity, y , is decreasing exponentially with respect to another, x , then

$$y = y_0 e^{-\lambda x}$$

Taking the natural logarithms on both sides of this relationship we get

$$\begin{aligned} \ln(y) &= \ln(y_0) + \ln(e^{-\lambda x}) \\ \ln(y) &= \ln(y_0) + (-\lambda)x \quad \dots\dots\dots (1) \\ \uparrow \quad \quad \uparrow \quad \quad \uparrow \quad \uparrow \\ y &= c \quad + m x \end{aligned}$$

We can therefore transform an exponential graph to a linear graph if we plot a graph of $\ln(y)$ as a function of x . Note the the gradient of the linear graph will be the negative of the decay constant, λ , that is $m = -\lambda$, and its y -intercept will be the natural logarithm of the starting value or $c = \ln(y_0)$.

Semi-logarithmic graph paper

To save the effort of first calculating the logarithms special graph paper called semi-logarithmic graph paper has been designed with the grid lines drawn at logarithmic intervals along one axis and linear divisions along the other. Such graph paper is used to draw exponential functions.
